

Skills Summary

Function Notation Is Not Multiplication

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading the notation as an instruction

Rule: in $f(x)$, the letter f is the function's *name* and the parentheses hold its *input*. There is no multiplication anywhere in the symbol.

So $f(4)$ says: wherever the rule shows the input variable, write 4 instead; then simplify. The name f never appears in the arithmetic.

Test yourself: if your work still contains the letter f , you multiplied instead of substituting.

Example: For $f(x) = 5x - 3$, find $f(4)$.

Step 1. The 4 sits inside the parentheses, so it is the input — this is a substitution, not a product.

Step 2. Replace x by 4: $5(4) - 3 = 20 - 3$.

$$f(4) = 17$$

Try it: For $f(x) = 6x - 1$, find $f(5)$.

$$\text{check: } 6(5) - 1 = 29$$

Skill 2 — Substituting a negative input

Rule: substitute the input *in parentheses*, then apply the exponents and signs. Writing -2 without parentheses is what turns a correct substitution back into a sign error.

Watch the square: $(-2)^2 = 4$, while $-2^2 = -4$. The first squares the input; the second squares 2 and then negates.

Example: For $h(x) = x^2 - 3x$, find $h(-2)$.

Step 1. The input is negative, so wrap it in parentheses before doing anything — that keeps the square and the subtraction honest.

Step 2. $(-2)^2 - 3(-2) = 4 + 6$.

$$h(-2) = 10$$

Try it: For $h(x) = x^2 - 3x$, find $h(-4)$.

$$\text{check: } h(-4) = 28$$

(!) Watch out: $f(a + b)$ is *not* $f(a) + f(b)$. The parentheses hold one input, so the whole sum goes into the rule at once. Check it on $f(x) = x^2$: $f(1 + 2) = 9$, but $f(1) + f(2) = 5$.

Skill 3 — Solving for the input

Two questions that look alike: “find $f(5)$ ” gives you the input and asks for the output — substitute. “Solve $f(x) = 5$ ” gives you the output and asks for the input — set the rule equal to 5 and solve.

Never divide by the function’s name. There is no factor to divide away; undo the rule one operation at a time instead.

Example: For $f(x) = 3x + 7$, solve $f(x) = 25$.

Step 1. The output is given and the input is unknown, so this is an equation to solve, not a substitution.

Step 2. $3x + 7 = 25$; subtract 7 to get $3x = 18$, then divide by 3.

$$x = 6$$

Try it: For $f(x) = 4x - 5$, solve $f(x) = 23$.

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